FOR EVERYONE — THE RESEARCHER, THE JOURNALIST, THE ONE WHO CHECKS

THE ABSENCE AUDIT — THE METHOD, AT NATIONAL SCALE

Tunisia — the complete program. Everything, in one document.

Five absences. Nine legs. One metabolism. Every number recomputed in code, every correction applied into the design, every claim graded.

Provenance law: MEASURED = cited anchor · DERIVED = arithmetic on measured values · ASSUMED = declared default. commercially-validated = 0 — no buyer has signed anything; published as zero on purpose. Generated 2026-09-09 by nations_audit.py / make_nations_pdfs.py — the same code that renders the page.

1. The five absences and the one metabolism

Five structural absences — monetary sovereignty (debt $40.5B, ~79% of GDP, 26.2% of export revenues to service), sovereign compute (InstaDeep: founded in Tunis, sold for $680M), strategic materials (phosphogypsum at Gabès costing the sea >$58M/yr), water, nutrition — answered by nine cash-positive legs on one metabolism of power, water, data and workforce. The audit is finished; the corrections are in the design.

2. The FX bridge — every dollar, by leg

Each leg below is volume × price × utilization, with the double-counting checks applied — the design carries the checked ledger.

LEG	TYPE	VOLUME	PRICE	$M/YR	GRADE
Strategic metals (U + heavy REE, acid stream)	gross export	716,500 lb U_3O_8 + 585 t REO	$80-95/lb U_3O_8 · $30/kg REO	$75-86M	**STRONG**
Solar fuel substitution (500 MW)	import substitution	745 GWh/yr displaced (876 GWh less 131 GWh to H_2)	$142/MWh (gas $12/MMBtu x 11 MMBtu/MWh + $10/MWh O&M); government anchor: 1,100 GWh ≈ $125M/yr gas saving for the same 500 MW fleet (Pg14)	$95-116M	**PLAUSIBLE**
Hydrogen → ammonia (loop-internal offtake)	import substitution	2.63 kt H_2/yr → 14.9 kt NH_3 for the GCT chain	$500/t imported ammonia displaced (external offtake upside: $8-9.5/kg H_2)	$7-25M	*SPECULATIVE*
Terfas (desert truffles)	gross export	940 t/yr mid (2,500 ha x 376 kg/ha; 70% exported)	EUR20-40/kg farm-gate	$14-29M	**PLAUSIBLE**

LEG	TYPE	VOLUME	PRICE	$M/YR	GRADE
Food import substitution	import substitution	1% of the $3.0B food import bill (frame; volume bridge is a pilot deliverable)	imported protein $15-25/kg vs local $5-10/kg	$30-30M	*SPECULATIVE*
Compute colocation services	gross export (services)	colo park, tenant-dependent	market colocation rates	$15-15M	*SPECULATIVE*
ESG financing delta	financing delta (not FX)	100-150bp on ~$800M portfolio debt	interest saved, not foreign exchange	$8-12M	**PLAUSIBLE**
Total external-account effect (mid)				**~$279M/yr**	**$80M strong · $137M plausible · $61M speculative · $0 verified**

By kind: exports $117M · import substitution $152M · financing delta $10M (interest saved, honestly labeled as not-FX). **By grade:** $80M strong · $137M plausible · $61M speculative · $0 verified — nothing is commercially validated yet, and that ledger reads zero on purpose.

3. The energy ledger — every kWh, by leg

LEG	GWH/YR	THE FORMULA
Pump-as-turbines (return flows)	3.9	70% × 150k m^3/d × 50 m × 75% × 8,760 h
Heat (kiln + exotherm)	30	9% of 333 GWh envelope
Flexibility (load shift)	20	dispatch value — not recovered joules
CO&sub2; mineralized	30 kt	mass, not energy
Total	**33.9 recovered + 20 dispatch**	**isobaric ERD inside the 4 kWh/m^3 benchmark — never double-counted**

4. Phase 0 — every dollar, itemized

ITEM	COST	WHAT IT BUYS
Water panel: 100 points, full suite (F, Cd, Pb, Cr, As, salinity, nitrate)	$35,000	the first anchor measurement
Acid-stream assays: 12 points (U, REE, Ra-226, Po-210, full metals) + GCT access	$40,000	the second anchor measurement
Emissions inventory: 30 passive samplers (PM2.5/SO2/fluorides) + lab	$25,000	baseline for the health scoreboard

ITEM	COST	WHAT IT BUYS
Cohort protocol: 3,000-family registry, ethics, data platform v0	$35,000	the DALY ledger becomes MEASURED
Health pilot first tranche: 100 households + 2 schools, 90 days	$58,000	filters, clean-air boxes, KPIs
Legal + finance: health fund, industrial SPV, land-lease pre-agreements, tender registration	$25,000	structure before scale
Engineering + project management	$35,000	the 90 days are run
List price	**$253,000**	**list price $253k; the two anchor measurements cost $75k and land first — the claimed range is the phased-execution plan, not the list**

5. Every claim, graded

CLAIM	GRADE	WHY
$40.5B debt (~79% GDP); 26.2% of export revenues to service	**STRONG**	MEASURED via World Bank API on 2026-09-09: debt $40.46B, debt/GDP 78.7%, service 26.17% of export revenues (2024)
$3.0B food and $4.9B fuel imports	**STRONG**	fuel $4.895B verified on WITS/Comtrade (HS27, 2024); food per the WB food classification
InstaDeep founded in Tunis, sold for $680M	**STRONG**	public deal record; the emblem for talent without infrastructure
Gabès: >$58M/yr of REE value lost to the sea	**STRONG**	Science of the Total Environment (2021): up to 1,523.67 t/yr REE, ~$58M/yr losses
U + heavy REE from the acid stream, ~$80M/yr	**STRONG**	flowsheet industrial since 1978 (Freeport), patent expired, Morocco building it now; volume derived from cited primitives; Gafsa-specific assays are Phase 0
~$285M/yr external-account effect, tiered	TIERED	$80M strong (proven flowsheet/cited anchor) · $137M plausible (gated) · $68M speculative (assumed volume/price)
33.9 GWh/yr recovered + 20 GWh/yr dispatch value	DERIVED	every leg computed from cited primitives; dispatch value honestly labeled — not joules
$2,592 per healthy year (water rung)	**PLAUSIBLE**	arithmetic DERIVED in code; DALY anchors transferred from Iranian studies — the cohort baseline makes it MEASURED
Phase 0 = $150-250k, itemized	**PLAUSIBLE**	itemized list price $253k; the two anchor measurements cost $75k and land first
Commercially validated	**ZERO**	no buyer has signed anything — published as zero on purpose

6. The hidden dependencies

DEPENDENCY	TYPE	STATUS	NOTE
Desal brine permit	regulatory	in design	Gabes outfall / diffuser; permit required before SWRO FID
PG chemistry (Ra-226 partitioning)	technical	Phase 0 assay	flowsheet keeps radium in gypsum (Freeport precedent); re-verify on Gafsa stream
Uranium handling	regulatory	gate	CNSTN licensing, NORM transport, export controls
REE separation capex	capital	later phase	+$244M solvent-extraction plant beyond the ~$100M circuit
Solar bankability	financial	gate	DSCR ~0.9-1.0 at the record-low tariff; bankable region or grant support required
Grid connection	infrastructure	later phase	3 GW connection + 400 kV upgrades ~$450M
Compute accelerators	regulatory	gate	export-control licenses; colocation until an anchor tenant is licensed
Truffle yield	agronomic	gate	376 kg/ha is the 20-yr Murcia average; gate = 3-season record >=200 kg/ha
Export prices	market	exposure	U spot $80-95/lb; truffle EUR20-40/kg — no contracted offtake yet
Financing	financial	uncommitted	DFI channel modeled for solar/desal; nothing committed

7. The structure and the pilot/capital separation

The design is six independently bankable modules sharing infrastructure. The loop is the outcome, never the precondition — no module's economics depends on another module succeeding, and every gate is public. World Bank CCDR (2023): Tunisia's water-sector investment need ~$17.1B and energy-sector ~$34.7B through 2050. The page is not that program and does not claim to be: Phase 0 is $150-250k of measurements, and the whole core build is ~$250-650M phased over years 2-5, module by module, each gated. The CCDR numbers confirm the scale of the problem; they do not compete with a pilot gate.

Africa Energy Portal (2026): 500 MW of utility-scale solar approved (~$400M investment). Same class as the page's P3 module (500 MW) — the record-low ~$31/MWh awards are the reference the DSCR gate is tested against.

8. The Fed transmission — measured, not restated

World Bank API, measured 2026-09-09 (Pg1): external debt **$40.46B**, **78.7% of GDP**; debt service **26.2% of export revenues**; reserves **$9.34B** = 4.3 months of imports; food $3.00B + fuel $4.90B of imports; CA -1.5% of GDP; growth 2.5%, inflation 5.2% (2025).

Channel 1 — the debt trap: the "original sin" (Pg12) — no long-term borrowing in the dinar abroad — exposes the state to the Fed cycle (Pg13): Fed hikes lift global yields, refinancing turns expensive, and the 26.2% of export revenues that debt service absorbs compounds while reserves (4.3 months) drain.

Channel 2 — imported inflation: the rate differential pulls capital to US safe assets, the dinar depreciates, and the dollar invoice on food ($3.00B) and fuel ($4.90B) inflates. The BCT — autonomous, non-monetizing under Organic Law 2016-35 (Pg4/Pg5) — must hold rates to defend the dinar: correct orthodoxy, strangled growth.

The loop's answer: owned export engines earn ~$117M/yr of gross exports outside the Fed cycle; the substitution legs take $152M/yr out of the dollar invoice directly; reserves rise, the rating runs B-/Caa1 → BB-, and the refinancing premium shrinks for the state with its own hard-currency engines. Not defiance of the Fed — independence from the channel.

The household ledger — what this does to the cost of living

A state program that does not make life cheaper for its people is decoration. Each line, computed from the same primitives as the rest of this document; the quarterly price index in Gafsa and Gabès is the published enforcement: **the price of protein and vegetables must bend downward within 24 months of the first harvest.**

HOUSEHOLD LINE	TODAY	AFTER THE LOOP
Water	$1.00-1.50/m³ (SONEDE produced)	$0.35/m³ coastal · $0.34/m³ inland
Protein	$15-25/kg imported	$5-10/kg local
Terfas	export €20-40/kg	one third stays home at ≤€12/kg, contractual
Health	the disease's bill	$2,592 per healthy year vs the $4,000/DALY standard
Energy (grid)	$0.110/kWh industrial	$0.031/kWh for the loop's loads; tariffs stay state policy
Work	28% unemployment (54% women graduates)	workshop, nursery, ponds, plants, crews

The energy ledger — what power costs, per party

Anchors cross-checked: Rw10 (0.352 TND/kWh industrial, 19 TWh), Pg14 (the government's own 1,100 GWh = 5% of production, ~$125M/yr gas saving), Rc13 ($31/MWh inside the $24-35 historical bid band), Pg15 (fuel imports $4.895B, WITS-verified).

PARTY	TODAY	AFTER THE LOOP	THE DELTA
The loop's industries	grid $0.110/kWh	**$0.031/kWh** on the loop's PPA	−72%
The state	19-22 TWh/yr, ~95% gas, $4.90B fuel bill	876 GWh/yr solar (4.0-4.6% of production)	−$106-125M/yr gas imports ($114-142/MWh band)
The household	subsidized STEG tariff	the cost stack beneath the tariff falls	the inflation channel throttled — every dinar protected

The frontier — where the loop reaches, and where it honestly stops

The complete household budget, swept — nothing hidden, nothing promised beyond the frontier.

HOUSEHOLD LINE	STATUS	THE HONEST NOTE
Electricity (household tariff)	**indirect**	household tariffs stay subsidized STEG policy; the loop lowers the cost stack they sit on and the subsidy burden the state carries
Cooking gas (the bonbonne)	**out of scope**	the loop displaces grid gas, not bottled LPG; the bonbonne subsidy is a separate state decision
Transport fuel	**out of scope today**	the hydrogen leg serves ammonia for the fertilizer chain; transport fuels are a later policy question
Wheat and bread subsidy	**out of scope**	the protein stack complements the grain program; it does not replace it
Water	**covered**	cost floor $1.00-1.50 -> $0.35 coastal / $0.34 inland
Health	**covered**	$2,592 per healthy year vs the $4,000 standard
Protein	**covered**	$5-10/kg local vs $15-25/kg imported
Work	**covered**	the workshop, nursery, ponds, plants, crews where unemployment is 28% (54% for women graduates)
Housing	**out of scope**	the loop does not build housing; named so it is not promised

The water stack — line-by-line, with each credit's failure mode

The $0.347/m³ audit: the base is measured anchors; each credit is a loop-internal yield with a named failure mode; the degraded cascade still clears the $0.68 gate.

LINE	$/M³	PROVENANCE	WHAT MAKES IT FAIL
Base — solar energy 4 kWh/m³ at the loop's own PPA + capex + O&M	$0.567	MEASURED anchors (Rc13/Rw2/Rw3)	holds — if even the PPA lapses, grid power takes the base to $0.88, still under SONEDE
DC reject heat → RO feed preheat — one seawater pipe cools the servers, then arrives warm at the membranes (SEC −2.5%/°C)	−$0.007	DERIVED — Leg 03 + Leg 01, same pipework	fails only if no compute anchor tenant — and costs $0.007
Kiln/exotherm preheat — the acid plant and board kilns warm the same feed	−$0.000	DERIVED — Leg 04/05 waste heat	negligible by design — carried at zero margin
Brine valorization — salt/Mg evaporated on the loop's own free heat, sold at the park gate (of-take signed; nothing hauled)	−$0.137	DERIVED — brine is the desal's own by-product	if the of-take is not signed by FID, the credit degrades to −$0.03 (disposal avoided only) — the biggest single sensitivity
Shared civil works — one intake, one outfall, one grid connection serves RO + DC + metals in the same park	−$0.032	DERIVED — 12% capex sharing	if the park phases one module at a time, halve it to −$0.016

LINE	$/M³	PROVENANCE	WHAT MAKES IT FAIL
Dispatch value — the desal's own flexibility follows the loop's own solar curve	−$0.015	DERIVED — 20 GWh shifted	fails to zero if the STEG contract does not reward flexibility — contractual, not hoped
ESG financing delta — the health leg's public KPIs reprice the loop's own debt (100–150bp)	−$0.026	DERIVED — nations_v3	fails to zero if lenders do not accept the KPI track record — bankable-region fallback keeps the module alive
Brine pump-as-turbine — the desal's own residual pressure	−$0.002	DERIVED — 2.5 bar × 75%	fails to zero — immaterial
Integrated	**$0.347/m³**	**28% of SONEDE's $1.00-1.50 (Pg16)**	**the degraded cascade: if every soft credit fails, water still lands at $0.521/m³ — under SONEDE's own $1.00–1.50, and the $0.68 gate still clears with margin. The credits are the margin, not the plan.**

The Darwinian ledger — why business as usual loses

Protein built the human brain (Pg27); the same lever, applied on purpose, is the plan's deepest one. The business-as-usual ledger is the fear-facts: what “keep doing what we do” actually buys.

FACT	THE NUMBER	WHAT IT MEANS FOR THE NEXT GENERATION'S BRAINS
Fluorosis belt	24% dental fluorosis (Rw1)	high-F water ↔ lower child IQ (Pg28) — BAU taxes the cortex before it forms
Protein poverty	first 1,000 days (Pg29)	permanent cognitive deficit — the trees version, built child by child
Brain drain	tertiary emigration among world's highest (Pg30)	the country exports its cortex like its phosphate
The proof	InstaDeep sold for $680M	the cognitive gold was real — and sold, not kept
Debt	26.2% of export revenues to service (Pg13)	the budget cannot buy the interventions the brain needs
The idle cohort	28% unemployment; 54% women graduates (Rm19)	built brains, no environment — adaptation = exit
Inflation	5.15% on $4.90B fuel + $3.00B food	the first cut is diet quality — the Fed reaches the children's IQ

SELECTION PRESSURE	WITHOUT THE PLAN	WITH THE PLAN
Water scarcity	selects emigration — the skilled leave first	the mesh re-specifies the environment; adaptation becomes staying
Protein poverty	caps cognition permanently	ponds + canteens + quota reverse the pressure
The debt cycle	the surplus is eaten; the economy cannot invest	owned exports + substitution cut the invoice
Institutions	gates stay closed; skills emigrate through them	public gates select for competence, in public

The open-field plan — terfas, hectare by hectare

RUNG	HECTARES	SEEDLINGS	PRODUCTION	VALUE	OPENS ON
Gate (yrs 1-4)	100	600,000	≥200 kg/ha	first fruit ≥€20/kg	3-season record + contract
Scale (yrs 4-8)	2,500	15M	940 t/yr	€16544-29704M/yr	nursery output
Ceiling (yrs 8+)	3,000	18M	1,128 t/yr	inside 1,000-1,500 t/yr market cap (Rc8)	market offtake signed

6,000 mycorrhized seedlings/ha (Rc16) — the nursery is the first asset: 5M seedlings/yr of local-ecotype capacity, ~200 nursery + ~1,000 field + ~783 seasonal jobs at scale. Local wild genetics, not the imported soil biology that failed in Saudi Arabia (Rc17).

The AI spine — the system's nervous system, and its boundary

The AI runs the measurement, the routing and the publication — never the decisions.

COMPONENT	WHAT IT DOES	WHAT IT IS WORTH
The dispatcher	routes every joule and m³ in real time: solar → desal → electrolyzer → ponds → DC cooling, following the solar curve minute by minute	worth the 20 GWh/yr of dispatch value — the AI is what collects it
The scoreboard	computes and publishes every gate number automatically — filter uptime, urinary F, DSCR, yield records — tamper-evident, no human can edit a gate	the public contract, machine-audited
The price index	tracks the Gafsa/Gabès household basket quarterly and publishes the bend — the people's enforcement metric, computed not curated	the number that proves the prices bend, or proves they do not
The cohort engine	runs the 3,000-family health baseline, dose-outcome analytics and anomaly flags for the clinical team	the DALY ledger becomes a live dataset
The dispatcher twin	a digital twin of the loop (energy, water, brine, heat) that simulates before any physical change and audits after	the falsification ladder, automated
The boundary	the AI never sets a gate, never signs a contract, never prices a tariff — it measures, routes and publishes. The decisions stay human, in public.	the guardrail that keeps the nervous system a servant, not a master

The results framework — what a World Bank board reads first

The scoreboard IS the M&E system (PforR class, Pg23).

INDICATOR	BASELINE	TARGET	DATA SOURCE
Filter uptime, pilot households	0% (filters not installed)	≥70%	scoreboard, live
Urinary fluoride, pilot children	baseline to be measured (Day 1-30)	−20%	cohort engine

INDICATOR	BASELINE	TARGET	DATA SOURCE
Household price index, Gafsa/Gabès	baseline Q0	bend downward ≤24 months after first harvest	price index, quarterly
Water produced cost	SONEDE \$1.00-1.50/m^3 (Pg16)	≤\$0.35/m^3 integrated	plant telemetry
Energy recovered inside the loop	0 GWh/yr	33.9 GWh/yr + 20 GWh dispatch	dispatcher ledger
Gas imports displaced	$4.895B fuel bill (Pg15)	−$106-125M/yr	WITS Comtrade, annual
Uranium recovered from the acid stream	0 (lost to the sea)	≥85% recovery gate	assay chain
Terfas hectares planted	0 ha	100 ha gate → 2,500-3,000 ha	nursery registry
Jobs in the basin	28% unemployment (54% women graduates)	workshop → nursery → plants → crews	employment registry

The people's energy is electrons, not hydrogen — and the H&sb2; lives inside the loop

The loop's H&sb2; (2.63 kt/yr) is an industrial molecule: per kWh of heat, green H&sb2; at \$0.21 is 7× the loop's \$0.031/kWh electron; blending caps at 5-20% by volume (Pg20/Pg21/Pg22); the whole H&sb2; volume equals 0.0% of grid-gas demand. The molecule goes where only a molecule works: ammonia for the GCT chain, inside the loop. the electrolyzer sits on the desal line, and everything it touches stays in the loop: the water it splits is the SWRO permeate (0.04% of the 150,000 m^3/d output); its un-split reject stream is clean water that returns to the product line; its oxygen by-product aerates the spirulina ponds and greenhouses; its stack heat (15–20% of input) joins the kiln and DC reject heat on the desal preheat circuit; and the minerals concentrated in its flush stream join the brine valorization line. The electrolyzer is not a consumer of the loop — it is another recovery point inside it.

9. The state program — architecture, instruments, financing, adoption

Counterparts — who signs what

INSTITUTION	INSTRUMENT	ROLE
GCT (Tunisian Chemical Group)	acid-stream MOU + sampling agreement	the treasury's feedstock
SONEDE	municipal/industrial water offtake at >=$0.68/m3 gate	water contracts under the existing regime
STEG	25-year PPA under the IPP regime (500 MW class)	the power leg's revenue
Ministry of Industry, Mines & Energy	program sponsor; tender registration	the executive mandate
Ministry of Health	biomonitoring + cohort registry	the scoreboard

INSTITUTION	INSTRUMENT	ROLE
CNSTN	NORM licensing for the uranium route	the regulatory pass
Ministry of State Domains & Agriculture	40-year leases under Law 93-120 (86,000 ha available)	the land
BCT	reserve/rating path; FX policy coordination	the monetary endgame

Legal instruments — nothing new needs to be invented

INSTRUMENT	WHAT IT PROVIDES	CITATION
Organic Law 2016-35 (BCT)	monetary autonomy; the program does NOT monetize	Pg4/Pg5
IPP regime — Law 2015-62 & 2024/25 awards	renewables concessions; ~$31/MWh PPAs; 500 MW precedent	Rc13
PPP framework — Law 2015-49	desal/metals as PPPs where the state is the offtaker	Pg2
Law 93-120 (state-domain leases)	40-year agricultural leases; ~86,000 ha earmarked	Rc19
NORM/radiological regime (CNSTN, IAEA standards)	the uranium route's license layer	Pg11/Rm12

Financing stack — staged, gated, modeled in code

STAGE	FINANCING	GATE / CONDITION
Phase 0 ($150-253k)	grant / founder / early mandate	no DFI needed; 90 days
Health pilot ($58k)	ESG/grant	the first scoreboard numbers
Metals plant (~$100M)	60% debt @7% + offtake security; MIGA	DSCR >=1.2 at contract prices (nations_v5)
Solar (500 MW, ~$550M)	DFI stack: EIB/AfDB/WB-IFC at 4.5% + MIGA; or lean build	DSCR gate inside the bankable region (nations_v5)
Desal (~$165M)	70% debt @6% + municipal offtake + sovereign backstop	blended >=$0.68/m3, >=60% pre-sold
Terfas + food ($7-35M)	concessional agri + founder	nursery first
EU Global Gateway (17 GW)	the national envelope this program plugs into	Pg7

Governance

- every gate is a public number — published, dated, re-checkable
- health fund separate from the industrial SPVs; one shared data platform
- quarterly scoreboard: water chemistry, filter uptime, cohort KPIs, price index in Gafsa/Gabes
- each pilot passes or dies alone, in public — the falsification ladder is the governance
- commercially-validated ledger published at zero until the first signature moves it

Adoption path — what the state signs, in order

1. **Month 0:** the state signs: GCT sampling MOU + Phase 0 mandate ($150-250k, 90 days, no construction)
2. **Day 90:** two anchor measurements land: the 100-point water panel and the acid-stream assays — every projection becomes a signed number
3. **Months 4-18:** pilots on public gates: acid pilot, 100 ha terfas + nursery, spirulina pond, solar bid, desal offtake, health KPIs
4. **Years 2-5:** build only what passed: metals, solar, desal, health at basin scale — module by module, each with its own financing
5. **Years 5-10+:** the full loop: municipal spine, compute colocation, hydrogen, brine valorization — the destination the original design described

Investor essentials — the FX bridge tiered, and the pathway economics

By grade: **$80M strong** · $137M plausible · $61M speculative · $0 validated — the zero is published on purpose, and each gate is the instrument that moves it. Pathway economics, computed in code (nations_v5.py):

PATHWAY	REVENUE	CAPEX	GROSS MARGIN	PAYBACK	DSCR
P2 metals	$75-86M/yr	~$100M	0-37%	6.3 yr (mid)	2.40 @60% debt
P3 solar	$27.2M/yr	~$0M	71%	n/a	0.46 mkt / 0.69 DFI — clearing PPA ≥$45652/MWh
P4 desal	$30.1M @$0.55 → $38.3M @$0.70	~$0M	gate-dependent	n/a	0.54 @$0.55 → 1.36 @$0.70 — clears

Engineer essentials — the numbers an EPC checks first

Energy: 33.9 GWh/yr recovered + 20 GWh dispatch — PAT 3.9, heat 30, CO&sb2; 30 kt mineralized. Water: SWRO 4 kWh/m³ lift floor 1.09 kWh/m³ per 400 m (Rw4); gate $0.68 blended, ≥60% pre-sold. Materials: Gafsa rock 322 mg/kg REE avg (Rm1); 60-85% to PG (Rm2); 85-90% of rock-U to the acid stream (Ru1), >95% recovery proven 1978-99 (Ru1); Ra-226 214-970 Bq/kg vs the 1,000 Bq/kg exemption (Rm11/Rm12); Phase 0 assays at 12 stream points.

Health essentials — the DALY arithmetic and the pilot

$2,592 per healthy year against the $4,000/DALY standard; 10 ha spirulina feeds 178,082 children at the RCT dose; basin needs 0.7 ha. Pilot: 100 households + 2 schools, 90 days, $58,000 — gates: filter uptime ≥70%, urinary F ↓20%, results public.

What this document is: a concept-level engineering design whose numbers are computed in code from sourced primitives — recomputed, corrected, and applied into the design. Free and public.

What it is not: a commissioned state program, a bank-grade feasibility study, an investment offer, or a claim that any of this has been built. Nothing here is advice to any government or any investor. Nothing replaces geological, radiological, geotechnical or EPC-level surveys. Verify every figure and citation before relying on any of it.

Works cited

1. [Rm1] Rare Earth Elements in Phosphate Ores and Industrial By-Products (Gafsa avg 322 mg/kg, peaks >1,700) — https://www.mdpi.com/2075-163X/15/12/1232...
2. [Rm2] Global phosphogypsum: generation, stockpiles and 60-85% REE partition to PG — https://www.mdpi.com/2071-1050/17/19/8828...
3. [Rm3] Gabès phosphate plants REE characterization: enrichments to 1,075 mg/kg; Ce 80-309 mg/kg; negligible Dy/Tb — https://www.researchgate.net/publication/352414505_Rare_earth_elements_characterization_associated_to_the_phos...
4. [Rm4] Characterization and Leaching Kinetics of REE from Phosphogypsum in HCl (1.65 mol/L; 65.6% ΣREE; 73.8% Y) — https://www.researchgate.net/publication/361001660_Characterization_and_Leaching_Kinetics_of_Rare_Earth_Elemen...
5. [Rm5] Recovering REE from phosphogypsum as a secondary resource: scale-up failure modes (filtration, silica gel, co-extraction) — https://www.chinanhd.com/recovering-rare-earth-elements-from-phosphogypsum-as-a-secondary-resource-pathway/...
6. [Rm6] PhosAgro pilot-scale REE recovery from apatite-derived PG (5 m³/h — igneous, richer feed) — https://cris.vtt.fi/en/publications/pilot-scale-recovery-of-rare-earths-and-scandium-from-phosphogyps/...
7. [Rm7] Rainbow Rare Earths Phalaborwa: 2.2 Mtpa PG, 29.1% NdPr, pilot proven, OPEX <$13/kg (exception, not the rule) — https://www.rainbowrareearths.com/wp-content/uploads/2022/03/21-08-04-Sprott-Initiation.pdf...
8. [Rm8] 2024-2026 oxide prices: NdPr $64-103/kg; Dy ~$340/kg; Tb ~$1,800-1,875/kg; La $3.70/kg; Y <$5-10/kg; Ce ~zero — https://en.institut-seltene-erden.de/aktuelle-preise-von-seltenen-erden/...
9. [Rm9] Mixed rare-earth concentrate discount vs separated oxides (30-50%) — https://pubs.acs.org/doi/pdf/10.1021/acssusresmgt.5c00353...
10. [Rm10] Aclara: solvent-extraction separation plant Class-5 CAPEX $244M, OPEX $12/kg REO — https://investingnews.com/aclara-announces-update-on-its-rare-earths-separation-project/...
11. [Rm11] Tunisian PG radiological characterization: Ra-226 214-970 Bq/kg; U-238 88.5; Pb-210 755.4; co-precipitation risk — https://www.researchgate.net/publication/24188010_Characterization_of_phosphogypsum_wastes_associated_with_pho...
12. [Rm12] IAEA / EU BSS: 1 Bq/g (1,000 Bq/kg) exemption threshold for U-238/Ra-226; TENORM classification — https://www.mdpi.com/2071-1050/17/8/3473...
13. [Rm13] NORM disposal options and costs ($15-420 per 55-gallon drum) — https://www.ogj.com/home/article/17230971/norm-disposal-options-costs-vary...
14. [Rm14] China's 85-90% share of global REE separation/refining capacity — https://www.adlittle.com/sites/default/files/reports/ADL%20The%20rare%20earth%20imperative%202025.pdf...
15. [Rm15] Breakeven economics for non-Chinese REE projects (~$77/kg NdPr-equivalent; PG-grade requires >$150/kg) — https://www.elibrary.imf.org/view/journals/001/2026/072/article-A001-en.xml...
16. [Rm16] Hydrochloric acid prices: EU $124-150/t (Q4 2025); NA ~$350/t; China ~$15/t — https://chemtradeasia.sg/en/market-insights/hydrochloric-acid-supply-report-2024-2025-buyer-lessons...
17. [Rm17] GCT throughput (~6.5 Mt phosphate rock/yr; >10 Mt/yr PG at peak; 'hundreds of millions of tons' discharged at Gabès) — http://www.gct.com.tn/fileadmin/user_upload/Downloads/Rapport_Annuels/GCT-ANNUAL-REPORT-2012-PART1.pdf...
18. [Rw1] Largest SWRO plants: Ras Al-Khair 1,036,000 m³/d; Taweelah 909,200 ($847M-1.2B); Sorek II 672,000; Rabigh 3 600,000 — https://ide-tech.com/en/project/sorek-b-desalination-plant/...
19. [Rw2] SWRO CAPEX $800-1,200 per m³/day for mega-plants (>100k m³/d) — https://supra-international.com/insights/seawater-reverse-osmosis-swro-desalination-technology-comprehensive-a...
20. [Rw3] LCOW at plant gate: $0.40-0.60/m³ (Jubail 3A $0.41; Sorek II bid $0.32-0.45); energy share 40-60% of OPEX — https://pmc.ncbi.nlm.nih.gov/articles/PMC8953854/...
21. [Rw4] Pumping physics: 400 m lift = 1.09 kWh/m³ at 100% eff; 1.45 at 75%; friction 0.5-1.5 kWh/m³ over 300 km; realistic 2.0-2.5 kWh/m³ — https://www.engineeringtoolbox.com/water-pumping-costs-d_1527.html...
22. [Rw5] Jubail-Buraydah IWTP: 587 km, 650,000 m³/d, $2.26B ($3.85M/km); twin-pipe scale $5-7M/km — https://www.meed.com/saudi-arabia-awards-22bn-water-transmission-contract...
23. [Rw6] Delivered inland desalinated cost $1.10-1.50/m³ (plant gate + transport + pumping energy) — https://investriyadh.ai/intelligence/water-scarcity-investment/...
24. [Rw7] Tunisian water tariffs: SONEDE 0.200-2.310 TND/m³; irrigation 0.060-0.160 TND/m³; groundwater 0.112-0.162 TND/m³ — https://www.sonede.com.tn/tout-savoir-sur-leau/la-facture-deau-methodes-de-paiement/...
25. [Rw8] International transfer precedents: GMMR $33B (2.5-6.5 Mm³/d, $0.28-0.62/m³, fossil water); Riyadh levelized transport $0.42/m³; <$0.20/m³ is the agronomic ceiling — https://www.mei.edu/publication/whats-next-libyas-great-man-made-river-project/...
26. [Rw9] 24/7 firming: 3.03 GW @20% CF = 0.606 GW avg (zero margin); 9.15 GWh/night; BESS round-trip 85% → 12-15 GWh; $110-125/kWh → ~$1.5B; solar+storage LCOE $100-130/MWh — https://www.energy-storage.news/battery-storage-system-prices-continue-to-fall-sharply-bnef-and-ember-reports-...

27. [Rw10] Tunisia grid: STEG 5.9 GW, ~19 TWh/yr, 95% gas-fired, renewables <5%, industrial tariff 0.352 TND/kWh; budget deficit 5-7% GDP, debt ~83% GDP — https://www.trade.gov/country-commercial-guides/tunisia-electrical-power-systems-and-renewable-energy...
28. [Rw11] Agrivoltaics: 20-40% CAPEX premium over utility PV (+$0.30/Wp); NREL range $1.20-2.60/Wp — https://www.mdpi.com/1996-1073/19/1/102...
29. [Rw12] Tunisia land law: foreign ownership of agricultural land prohibited; 40-year leases only; state-domain conversion is an administrative hurdle — https://2009-2017.state.gov/e/eb/ifd/2008/101021.htm...
30. [Rw13] Grid constraints: 3 GW overwhelms local 400 kV; new lines $1-2.5M/km; renewables <5% of generation; curtailment risk — https://reglobal.org/tunisia-focuses-on-grid-expansion-for-integrating-renewable-energy/...
31. [Rc1] Underground DC precedents: Lefdal Mine (olivine; 200 MW; PUE 1.08-1.15 via 8°C fjord water); SubTropolis limestone (18°C + mechanical cooling); no phosphate-mine DC exists — https://www.lefdalmine.com/...
32. [Rc2] Gafsa basin: DInSAR-documented subsidence at Moulares; geothermal gradient 25-30°C/km → 23-29°C at 100-300 m depth — https://www.researchgate.net/publication/251701016_Phosphate_mine_subsidences_deduced_from_differential_interf...
33. [Rc3] Gafsa radiation: ^{226}Ra 16.3-375.1 Bq/kg in soils; phosphate tailings to 7,606 Bq/kg; radon-222 accumulation in enclosed workings; alpha-induced soft errors — https://www.researchgate.net/publication/380540277_Assessment_of_radiation_exposure_in_the_phosphate_mining_ar...
34. [Rc4] ASHRAE TC9.9 Class H1 (5th ed.): 18-22°C recommended, 25°C max for AI accelerators; AI datacenter CAPEX $20-35M/MW (incl. IT), $8-15M/MW facility-only — https://envigilance.com/compliance/ashrae-tc-9-9/...
35. [Rc5] GPU cost: ~100,000 B200-class per 100 MW; $25-40k each → $2.5-4B per 100 MW; 300 MW ≈ $7.5-12B — https://intuitionlabs.ai/articles/data-center-gpu-pricing-2026...
36. [Rc6] US BIS export controls 2024-26: UAE 35,000 GPUs (G42/HUMAIN, 'Compute Diplomacy'); Saudi case-by-case 35,000 cap; Tunisia has no bilateral AI-security agreement — https://www.commerce.gov/news/press-releases/2025/11/statement-uae-and-saudi-chip-exports...
37. [Rc7] Terfezia cultivation: commercial since 1999-2001 (Murcia, Spain); fruiting at year 2-3; 20-yr studies: average ~376 kg/ha, peak 1,069 kg/ha, drought years 0-2 kg/ha — https://pmc.ncbi.nlm.nih.gov/articles/PMC8907101/...
38. [Rc8] Desert-truffle water 200-250 mm/yr; protein 14-22.5% dry weight (the page's '20-27%' sits at the top of the reported range) — https://www.mdpi.com/2073-4395/11/8/1462...
39. [Rc9] Terfezia farm-gate €20-60/kg (€200/kg only in extreme scarcity in Gulf markets); Tuber melanosporum €700-1,500/kg retail — different commodities — https://trufamania.com/desert-truffles.htm...
40. [Rc10] Mycoculture economics: black truffle payback 10-15 years; Terfezia fruits faster but at one tenth the price — https://rovirosafarms.com/melanosporum/...
41. [Rc11] Tunisia sovereign ratings: Moody's Caa1 (Feb 2025 upgrade, stable); Fitch B- (Sep 2025, affirmed Jan 2026); reserves ~4.5 months import cover; WACC 10-15% — https://www.fitchratings.com/research/sovereigns/fitch-upgrades-tunisia-to-b-outlook-stable-12-09-2025...
42. [Rc12] Who finances Tunisia: EIB, AfDB, KfW, World Bank/IFC, EU Global Gateway (17 GW renewable programme); no active IMF program — https://international-partnerships.ec.europa.eu/policies/global-gateway/17-gw-renewable-energy-programme-tunis...
43. [Rc13] Empirical IPP record: 500 MW solar awarded 2024/25 (Scatec, Qair, Voltalia 25-yr PPAs at ~$0.031/kWh); installed renewables ~400 MW; STEG debt >$1.3B — https://www.engineeringnews.co.za/article/scatec-awarded-ppas-for-wind-solar-projects-in-tunisia-2026-01-26...
44. [Rc14] Chain logic: 8 interdependent stages at a generous 70% individual success ⇒ ~5.8% joint probability (0.7^8) — https://www.mdpi.com/2073-4395/14/12/3021...
45. [Rc15] Terfezia claveryi salt sensitivity: establishes only in near-fresh soils, EC 0.82-0.85 dS/m — https://jabonline.in/abstract.php?article_id=1363&sts=2...
46. [Rc16] Terfezia plantation density: ~6,000 mycorrhized plants per hectare — https://www.researchgate.net/publication/299679495_Preparation_and_Maintenance_of_Both_Man-Planted_and_Wild_Pl...
47. [Rc17] Saudi Al-Khalidiah 150-ha Tirmania plantation: largely failed (14 kg/ha after soil import; rainfall collapse) — https://ojs.openagrar.de/index.php/JABFQ/article/download/2184/2570...
48. [Rc18] Desert-truffle market depth: Mamora (Morocco) wild harvest peaks ~100 t/yr; luxury band breaks past ~1,000-1,500 t/yr of new supply — https://www.mdpi.com/1999-4907/14/5/952...
49. [Rc19] Tunisia arid land: ~86,000 ha of state domain earmarked for 40-year agricultural leases (Law 93-120) — https://documents1.worldbank.org/curated/en/926441468778187741/pdf/multi0page.pdf...
50. [Ru1] Uranium from wet-process phosphoric acid: Freeport (US) produced 14M lbs U_3O_8 1978-99 at >95% DEHPA/TOPO recovery; ~20% of US uranium in the 1980s came from phosphates — https://www.osti.gov/servlets/purl/1357599...
51. [Ru2] URANEXT/OCP (Morocco) building the first modern yellowcake-from-phosphoric-acid plant, ~$100M CAPEX, El Jadida, 2025 — https://northafricapost.com/87963-uranext-to-build-moroccos-first-yellowcake-uranium-plant-in-el-jadida.html...
52. [Ru3] WPA uranium SX cost $59-83/lb U_3O_8 (IX $46-76) vs $80-95/lb 2024-26 spot — https://repositories.lib.utexas.edu/bitstreams/89897275-85cf-4cd6-a0c5-1d8daa353799/download...
53. [Ru4] REE from phosphoric acid: PhosAgro pilot; D2EHPA extracts >85% of heavy REEs (Dy, Y, Er) from WPA — https://www.mdpi.com/2673-4117/7/2/58...
54. [Ru5] Gafsa phosphate rock uranium content 45-140 ppm (southern basin) — https://www.researchgate.net/publication/380540277_Assessment_of_radiation_exposure_in_the_phosphate_mining_ar...
55. [H1] Guissouma 2017 (Chemosphere): Tunisia tap-water fluoride 0-2.4 mg/L; 25% of population at dental-fluorosis risk, 20% at skeletal-fluorosis risk (WHO limits) — https://pubmed.ncbi.nlm.nih.gov/28284958/...

56. [H2] Bouchiba 2026 (Toxics 14(5):438): Mdhilla beneficiation effluent Cd 0.49, Pb 0.71, Cr 3.09 mg/L — exceeding discharge limits; risk driven by Cd, Co, Tl — https://doi.org/10.3390/toxics14050438...
57. [H3] Gupta 1996 RCT: fluorosis REVERSED in children (Ca + VitD3 + ascorbic acid; n=25; water 4.5 ppm F; double-blind) — https://pubmed.ncbi.nlm.nih.gov/8942013/...
58. [H4] Ghaemi 2024 (Iran): dental-fluorosis DALY rate up to 981.45 (UI 353.23-1618.40) per 100,000 in a high-F district; Monte Carlo dose-response — https://pubmed.ncbi.nlm.nih.gov/39003868/...
59. [H5] Abtahi 2019 (Water Research): age-sex specific DALYs from drinking-water fluoride, Iran national study — https://pubmed.ncbi.nlm.nih.gov/30953859/...
60. [H6] Naddafi 2022 (Env Research): burden of disease from heavy metals in drinking water, Iran 2019 — https://pubmed.ncbi.nlm.nih.gov/34529973/...
61. [H7] Egner 2014 RCT (n=291): broccoli-sprout beverage 12 wks -> urinary benzene mercapturic acid +61%, acrolein +23% (P<=0.01); GSTT1-positive excrete more — https://pubmed.ncbi.nlm.nih.gov/24913818/...
62. [H8] Pabis 2018: zinc supplementation -> >30% reduction of kidney/liver Cd in aged metallothionein-transgenic mice — https://pubmed.ncbi.nlm.nih.gov/30103140/...
63. [H9] Zhai 2016: Lactobacillus plantarum CCFM8610 binds gut Cd, protects the intestinal barrier, lowers tissue Cd in mice — https://pubmed.ncbi.nlm.nih.gov/27208136/...
64. [H10] Mueller 2018: N95-class respirators >=98% filtration efficiency vs <=44% for any cloth — https://pubmed.ncbi.nlm.nih.gov/29779694/...
65. [H11] Allen & Barn 2020: HEPA purifiers substantially cut indoor PM2.5 and improve subclinical cardiopulmonary health; population benefits often exceed costs — https://pubmed.ncbi.nlm.nih.gov/33241434/...
66. [H12] Drieling 2022 RCT (75 asthmatic children): HEPA in bedroom+living room -> poorly-controlled asthma IRR 0.43-0.45; unplanned clinical utilization IRR 0.35 — https://pubmed.ncbi.nlm.nih.gov/34980119/...
67. [H13] Spirulina RCT (Cambodia, n=179): 2 g/day x 50 days -> anemia reduction significant (p=0.004); weight-gain trend p=0.07 (do not overclaim growth) — https://pubmed.ncbi.nlm.nih.gov/36476193/...
68. [H14] EJAtlas: Gafsa affected population 30,000-70,000; cancer/asthma/infertility reported high; conflict since 2008 — https://ejatlas.org/conflict/phosphate-mining-in-gafsa...
69. [H15] NRGI: 'serious dearth of rigorous studies' linking phosphate operations to health outcomes; 7% birth-defect claim is activist-reported, unverified — https://resourcegovernance.org/articles/uncovering-impacts-phosphate-mining-tunisian-women...
70. [H16] Geographical 2024: asthma onset age 2 documented (Redeyef); nearest hospital 70 km; doctors prescribe leaving Gafsa — https://geographical.co.uk/science-environment/gafsas-silent-suffering-the-hidden-cost-of-phosphate-mining-in-...